

Python Programming Masterclass

LearnSphere Official Notes

Chapter 1: Basics of Python

Python is an interpreted, high-level and general-purpose programming language. Python's design philosophy emphasizes code readability with its notable use of significant indentation.

Your First Program:

```
# This program prints Hello, world!  
print('Hello, world!')
```

Chapter 2: Data Structures

Python has several built-in data structures such as Lists, Dictionaries, Tuples, and Sets. Understanding these is crucial for effective programming.

Lists & Dictionaries Example:

```
# A list of fruits
fruits = ["apple", "banana", "cherry"]
fruits.append("orange")
print(fruits[1]) # Output: banana

# A dictionary tracking student grades
student_grades = {
    "Alice": 85,
    "Bob": 92,
    "Charlie": 78
}
print(student_grades["Bob"]) # Output: 92
```

Chapter 3: Control Flow & Functions

Control flow allows you to dictate the order in which statements are executed in a script based on conditions.

Functions and Loops Example:

```
def greet_user(name):  
    """A simple function to greet the user."""  
    return f"Hello, {name}!"  
  
# Using a for loop with the function  
names = ["Alice", "Bob", "Charlie"]  
for name in names:  
    print(greet_user(name))  
  
# Output:  
# Hello, Alice!  
# Hello, Bob!  
# Hello, Charlie!
```

Happy Coding with Python!